

例. 一衍射光栅, 每厘米 200 条透光缝, 每条透光缝宽为 $a = 2 \times 10^{-3} \text{ cm}$, 在光栅后放一焦距 $f = 1 \text{ m}$ 的凸透镜, 现以 $\lambda = 600 \text{ nm}$ 的单色平行光垂直照射光栅; 求:

- (1) 透光缝 a 的单缝衍射中央明条纹宽度为多少?
- (2) 在该宽度内, 有几个光栅衍射主极大?

解: (1) 单缝衍射 暗纹 $a \sin \theta_1 = k\lambda = \lambda$

$$\sin \theta_1 = \frac{\lambda}{a} = \frac{600 \times 10^{-9}}{2 \times 10^{-5}} = 0.03 \quad \theta_1 = 0.03 \text{ rad}$$

$$x_1 = f \cdot \tan \theta_1 = 0.03 \text{ m} \quad d = \frac{1 \times 10^{-2}}{200} = 5 \times 10^{-5} \text{ m}$$

$\therefore$ 中央明纹宽度为 $\Delta x = 2x_1 = 0.06 \text{ m}$

(2) 光栅衍射主极大 $d \sin \theta = k'\lambda$

$$k' = \frac{d \sin \theta_1}{\lambda} = \frac{5 \times 10^{-5} \times 0.03}{600 \times 10^{-9}} = 2.5 \quad \text{取 } k'=2, \text{ 有 } k' = 0, \pm 1, \pm 2$$

共 5 个主极大

讨论: (1) $a \sin \theta = k \lambda$ $\tan \theta = x/f$

当 $x \ll f$ 时, $\tan \theta \approx \sin \theta \approx \theta$, $ax/f = k \lambda$, 取 $k=1$ 有

$$x = fl/a = 0.03 \text{ m}$$

$\therefore$ 中央明纹宽度为 $\Delta x = 2x = 0.06 \text{ m}$

(2) $(a+b) \sin \theta = k' \lambda$

$$k' = (a+b)x/(f \lambda) = 2.5$$

取 $k' = 2$, 有 $k' = 0, \pm 1, \pm 2$ 共 5 个主极大